

Using Ansible with z/OSMF





THE LAW OF LEAKY ABSTRACTIONS

The law of leaky abstractions: **"All non-trivial abstractions, to some degree, are leaky."**



In 2002, software engineer Joel Spolsky observed that abstractions are *leaky*. He meant that at times, details of the underlying layer leak through the interface and become visible to users of the abstraction.

If your knowledge of a system is limited to the view provided by the abstraction, there will be situations in which you experience difficulty troubleshooting.

His article uses TCP as an example. TCP is designed to be a reliable protocol, but it is built on top of IP, which is an unreliable protocol. He writes that TCP is "what computer scientists like to call an abstraction: a simplification of something much more complicated that is going on under the covers."

z/OSMF is a non-trivial abstraction that hides much of the complexity of what is happening on a z/OS instance. Ansible is a non-trivial abstraction that hides much of the complexity of what is happening in z/OSMF, and z/OS as well. There will be situations when the error messages that make it back to the control node are confusing or incomplete, and you must look at the z/OS environment to figure out what's happening.

TERRAFORM → ANSIBLE → Z/OSMF → WRAPPED APIS → Z/OS

Adding more and more layers of abstraction over something may or may not be a cost-effective and sustainable way to improve consistency, reliability, and productivity in managing numerous systems. It depends on how the organization chooses to manage such work, and on their strategy for evolving the technical environment going forward.

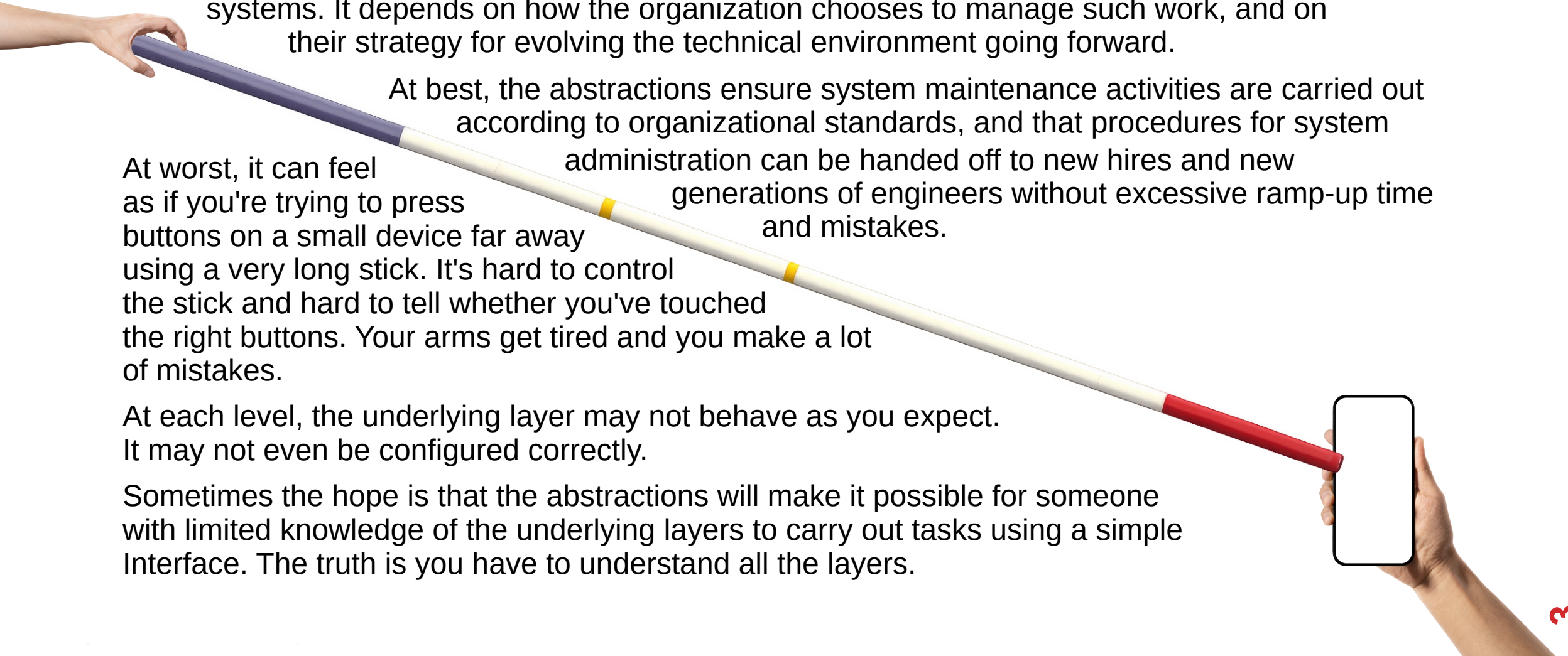
At best, the abstractions ensure system maintenance activities are carried out according to organizational standards, and that procedures for system

administration can be handed off to new hires and new generations of engineers without excessive ramp-up time and mistakes.

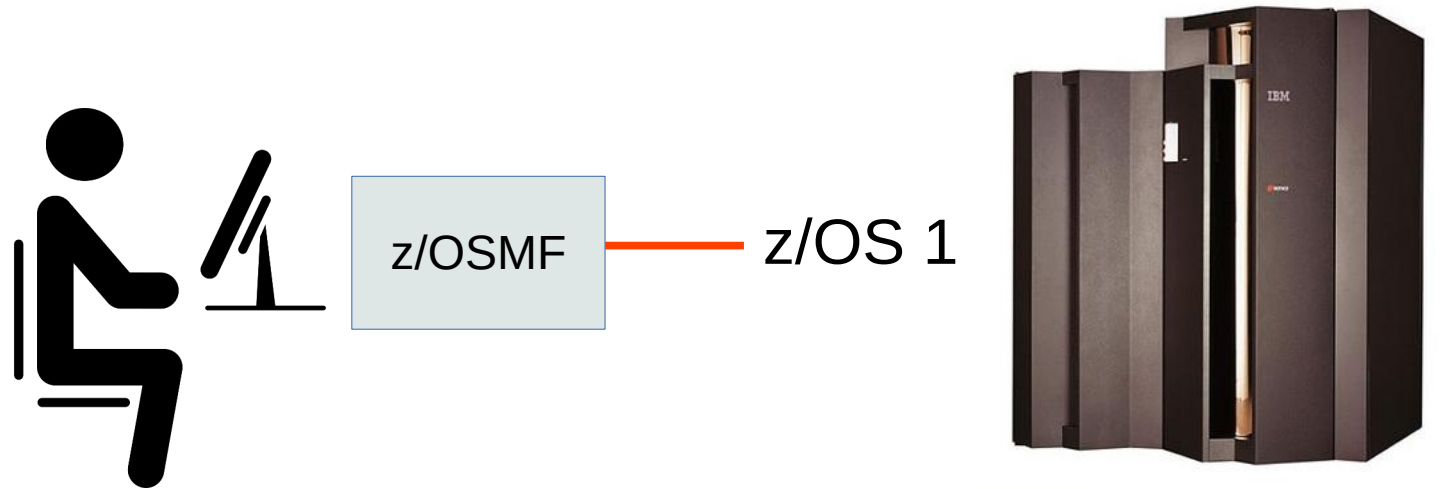
At worst, it can feel as if you're trying to press buttons on a small device far away using a very long stick. It's hard to control the stick and hard to tell whether you've touched the right buttons. Your arms get tired and you make a lot of mistakes.

At each level, the underlying layer may not behave as you expect. It may not even be configured correctly.

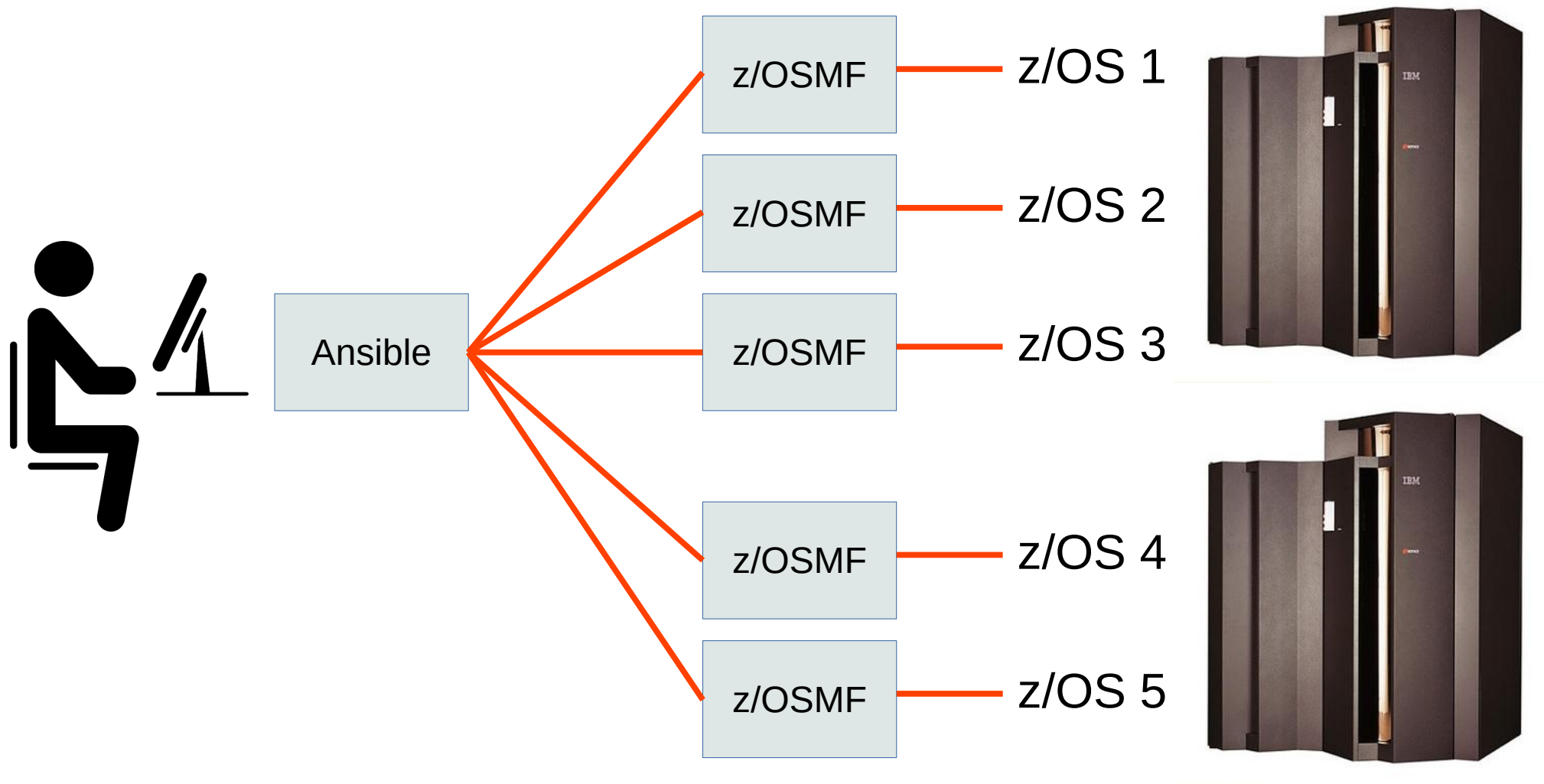
Sometimes the hope is that the abstractions will make it possible for someone with limited knowledge of the underlying layers to carry out tasks using a simple Interface. The truth is you have to understand all the layers.



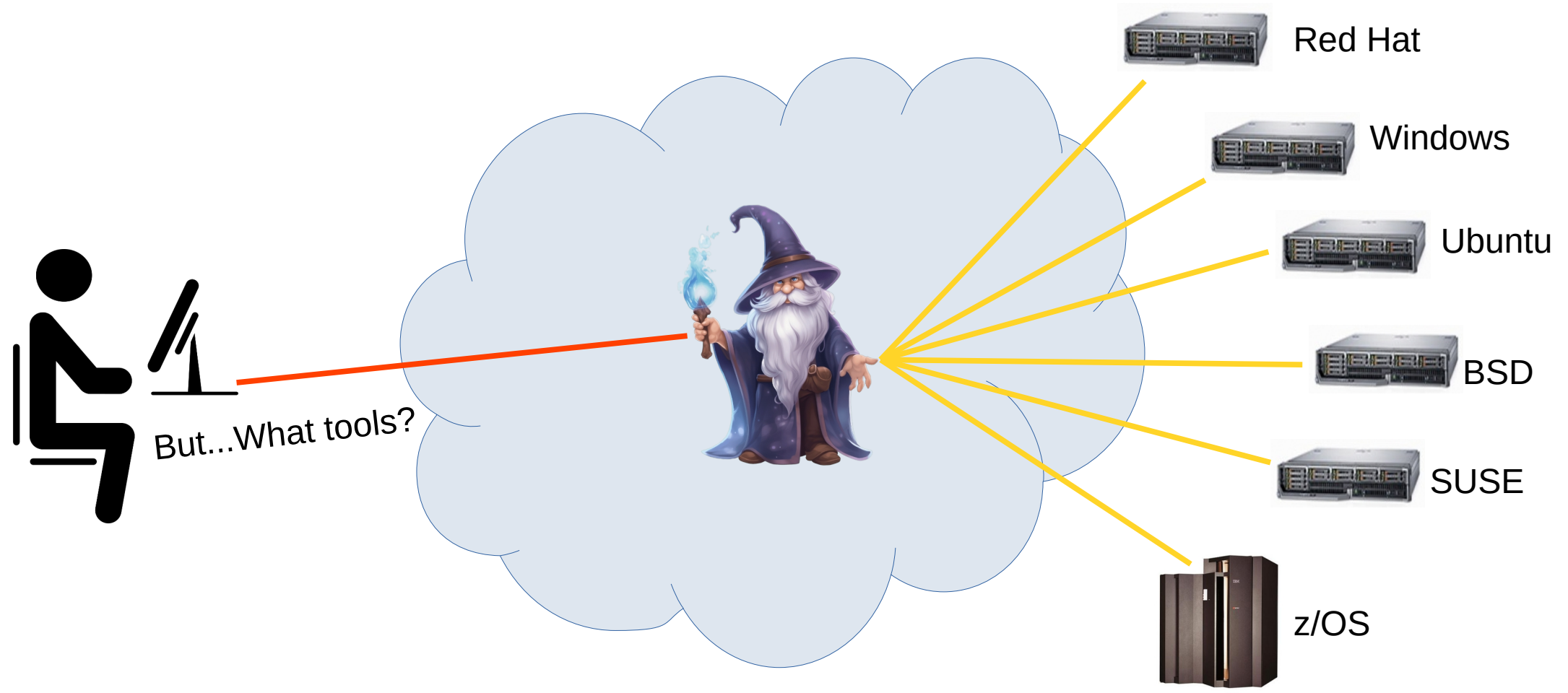
ADVANTAGE OF ANSIBLE PLUS Z/OSMF



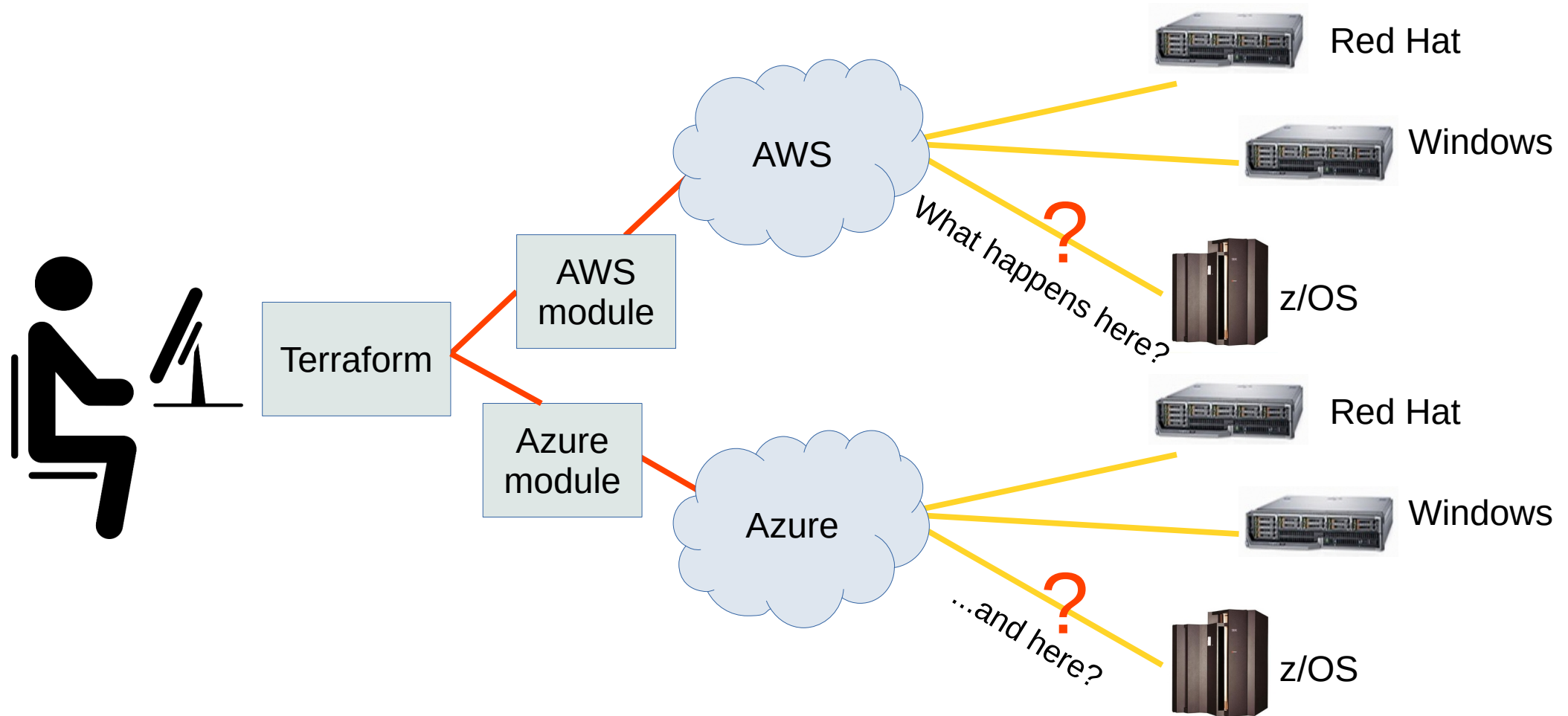
ADVANTAGE OF ANSIBLE PLUS Z/OSMF



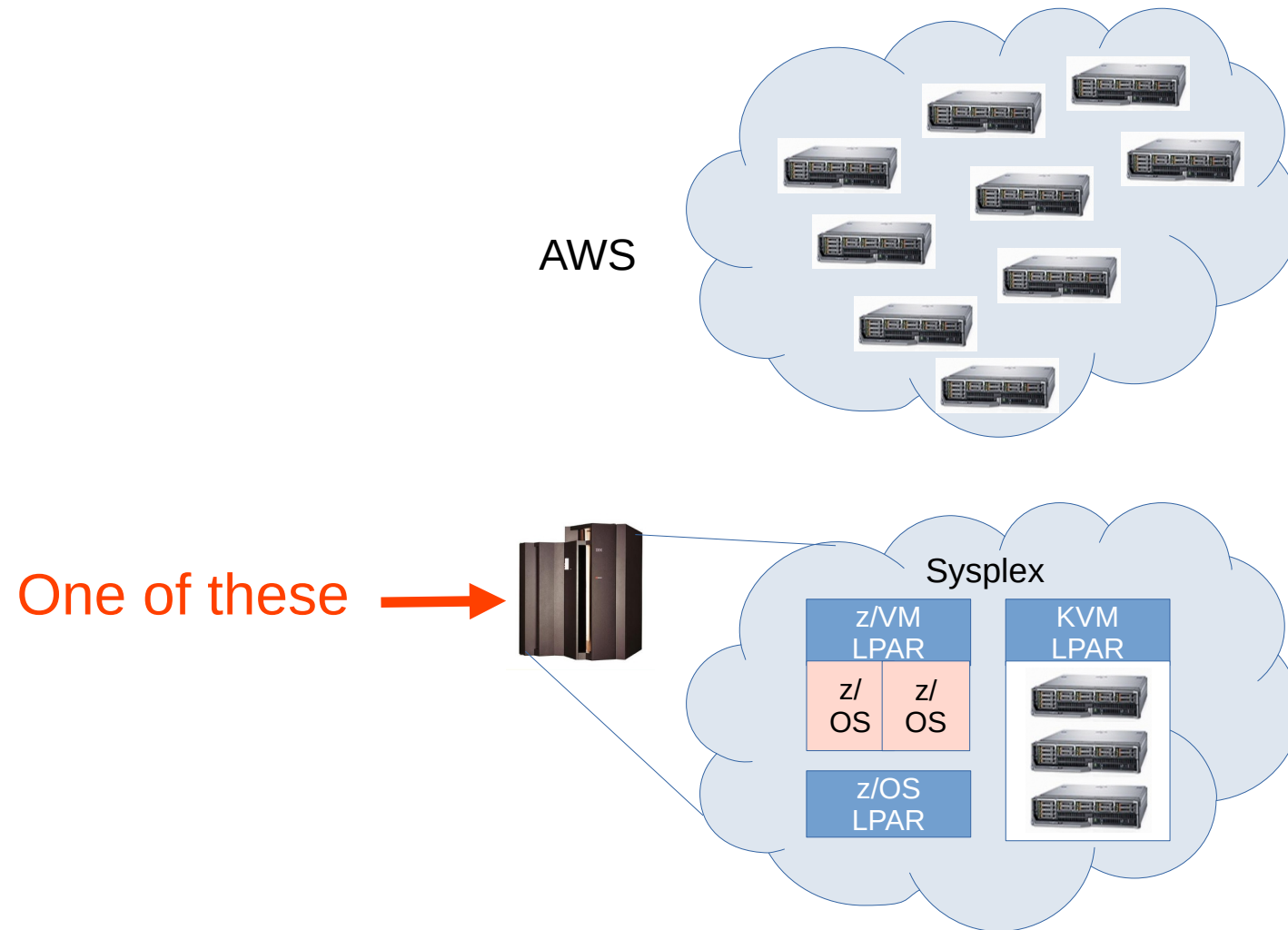
THE DREAM



THE WISH

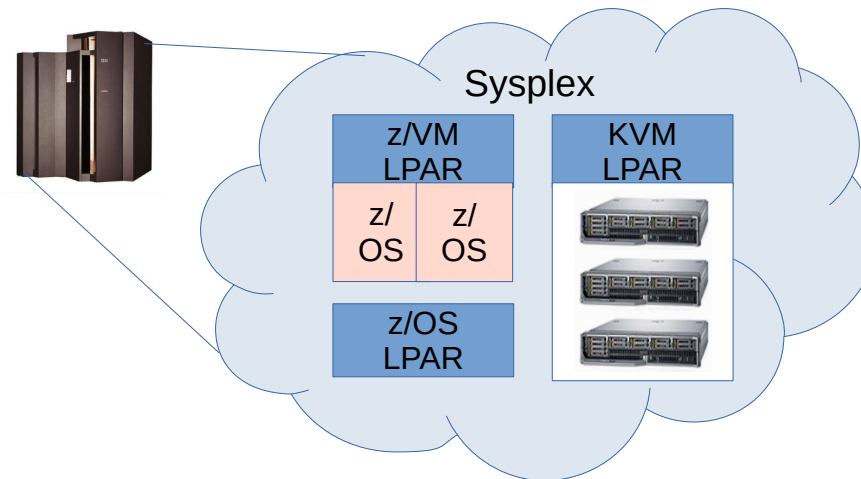
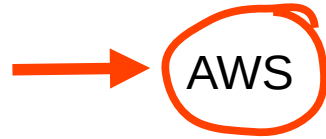


THE REALITY

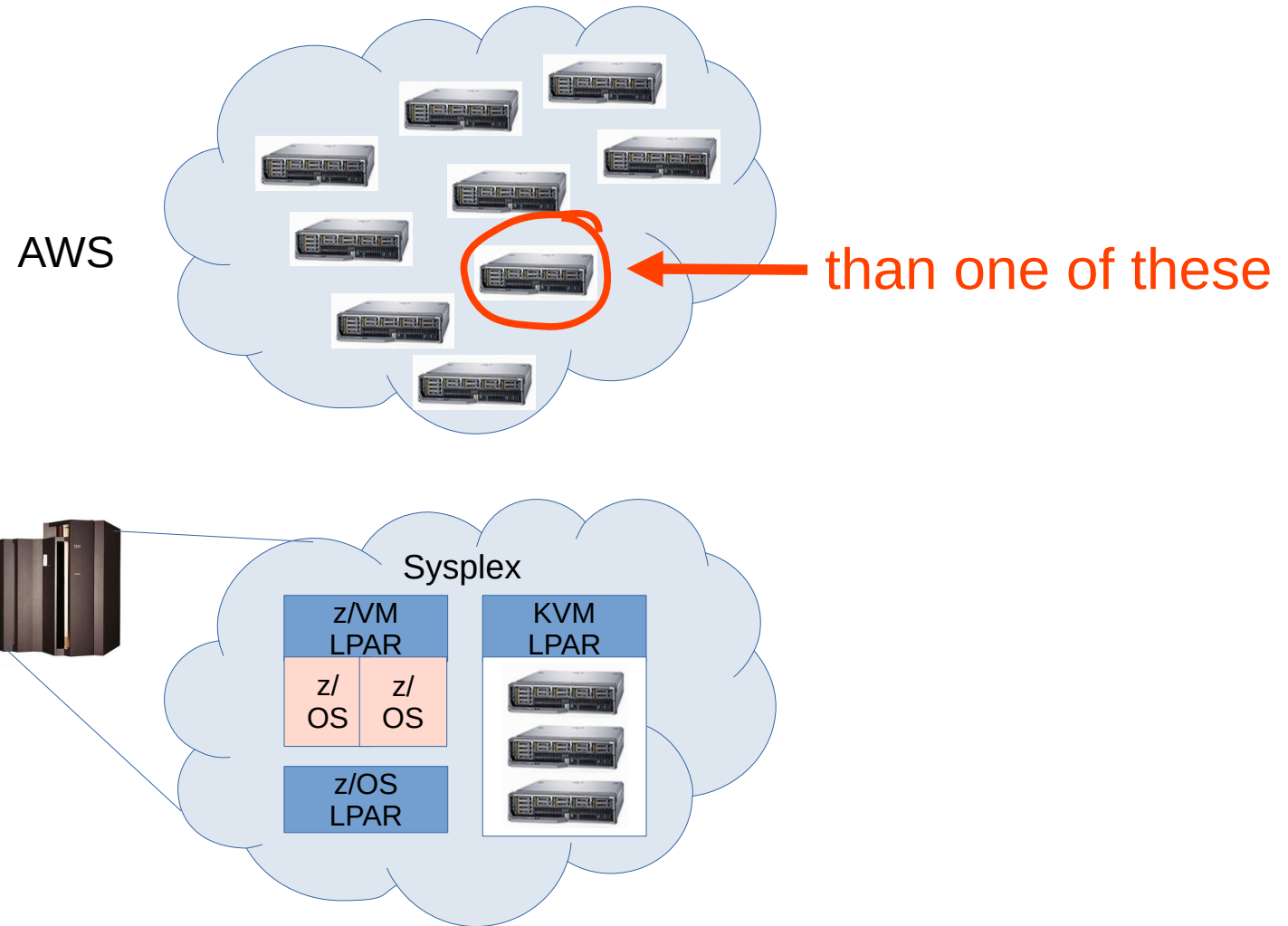


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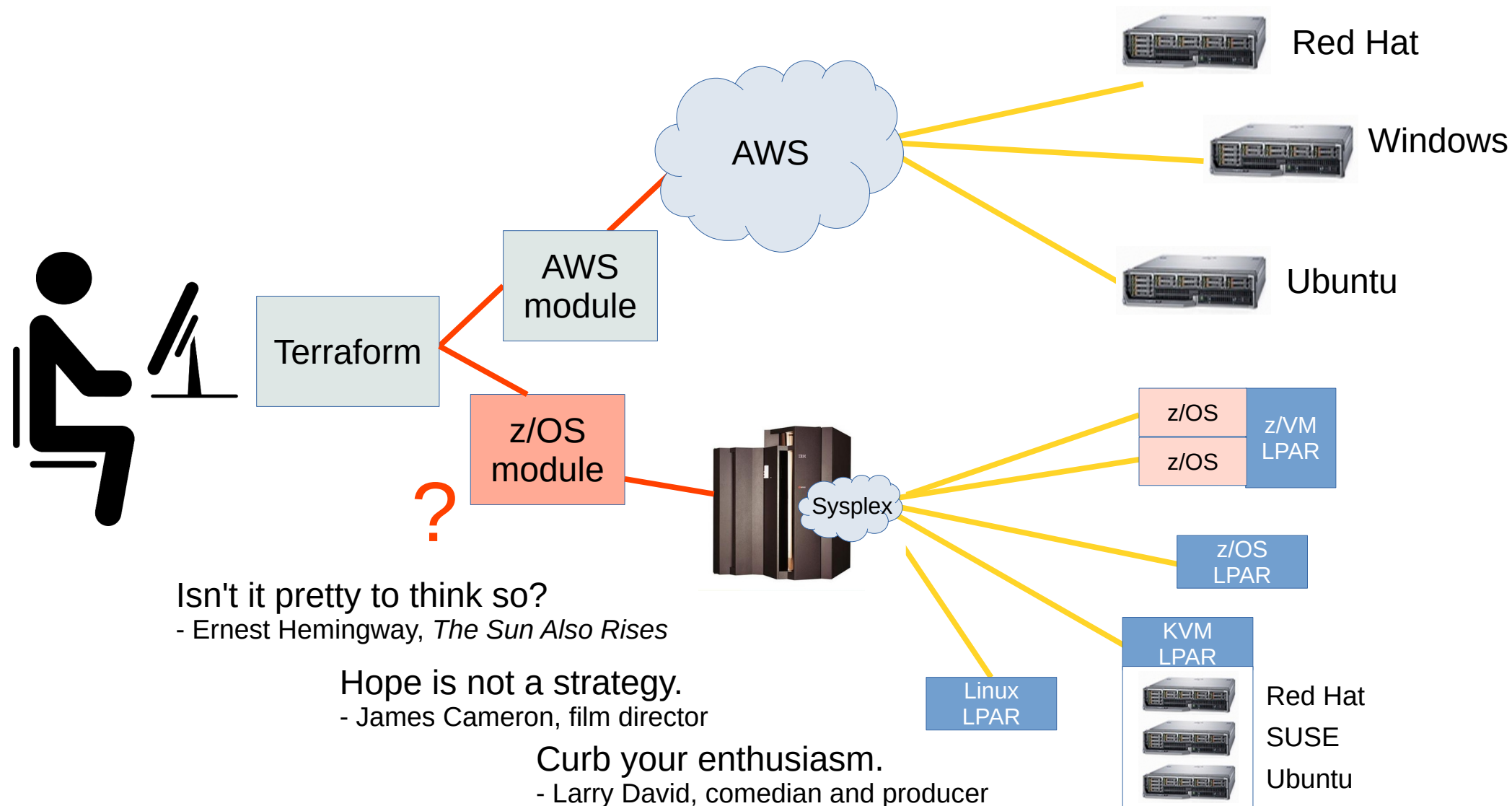
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one of these



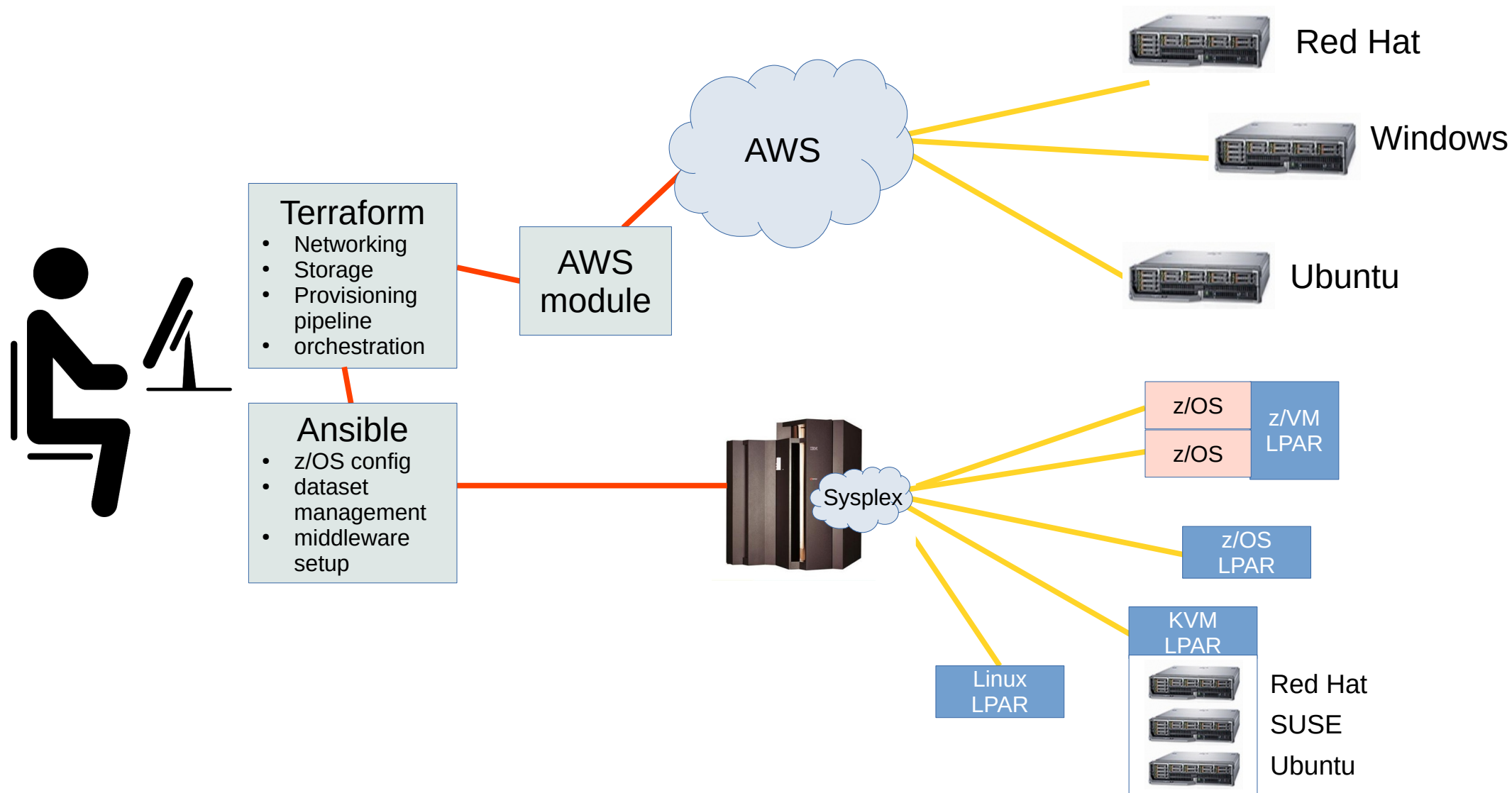
THE REALITY



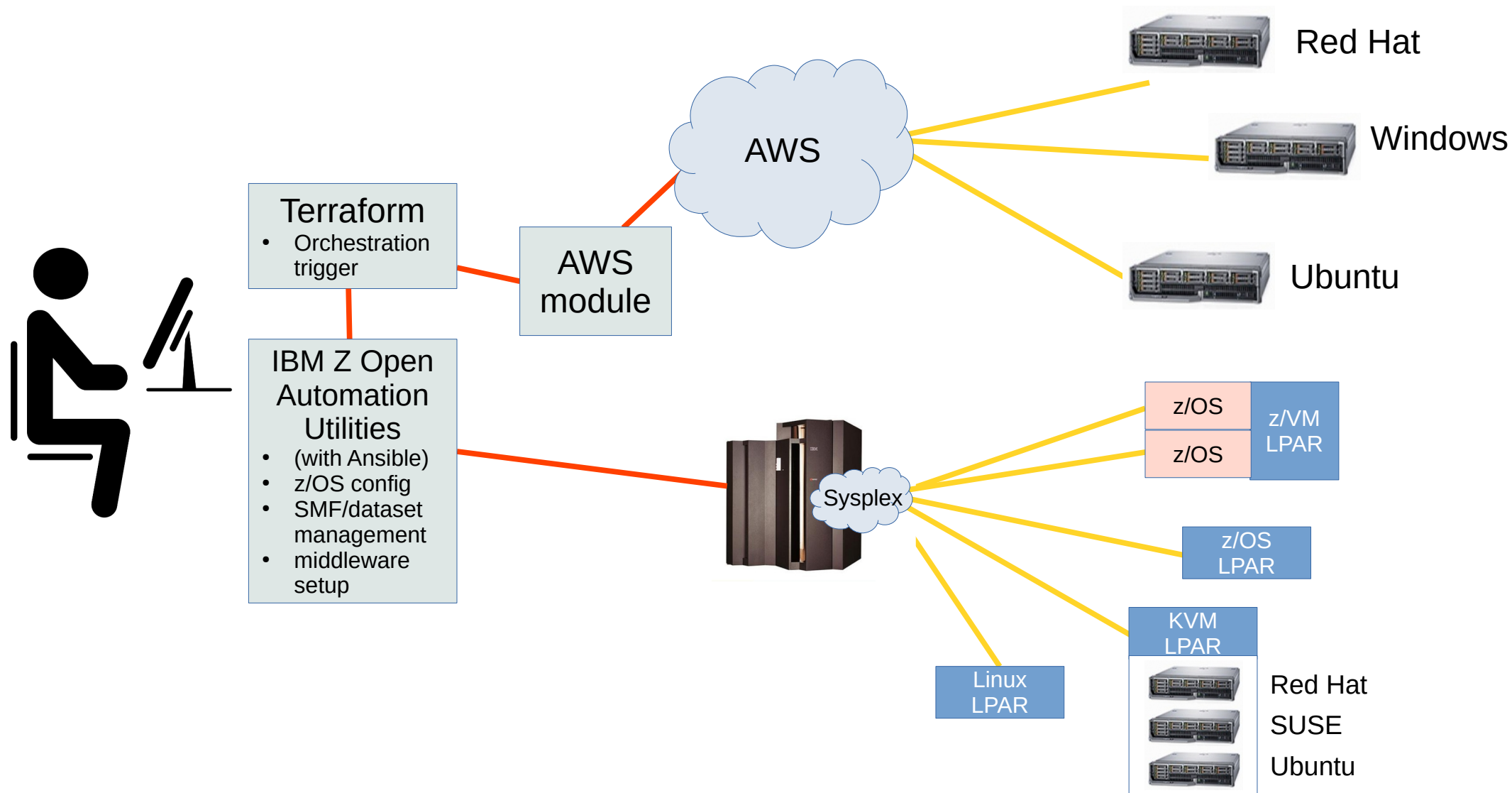
THE HOPE



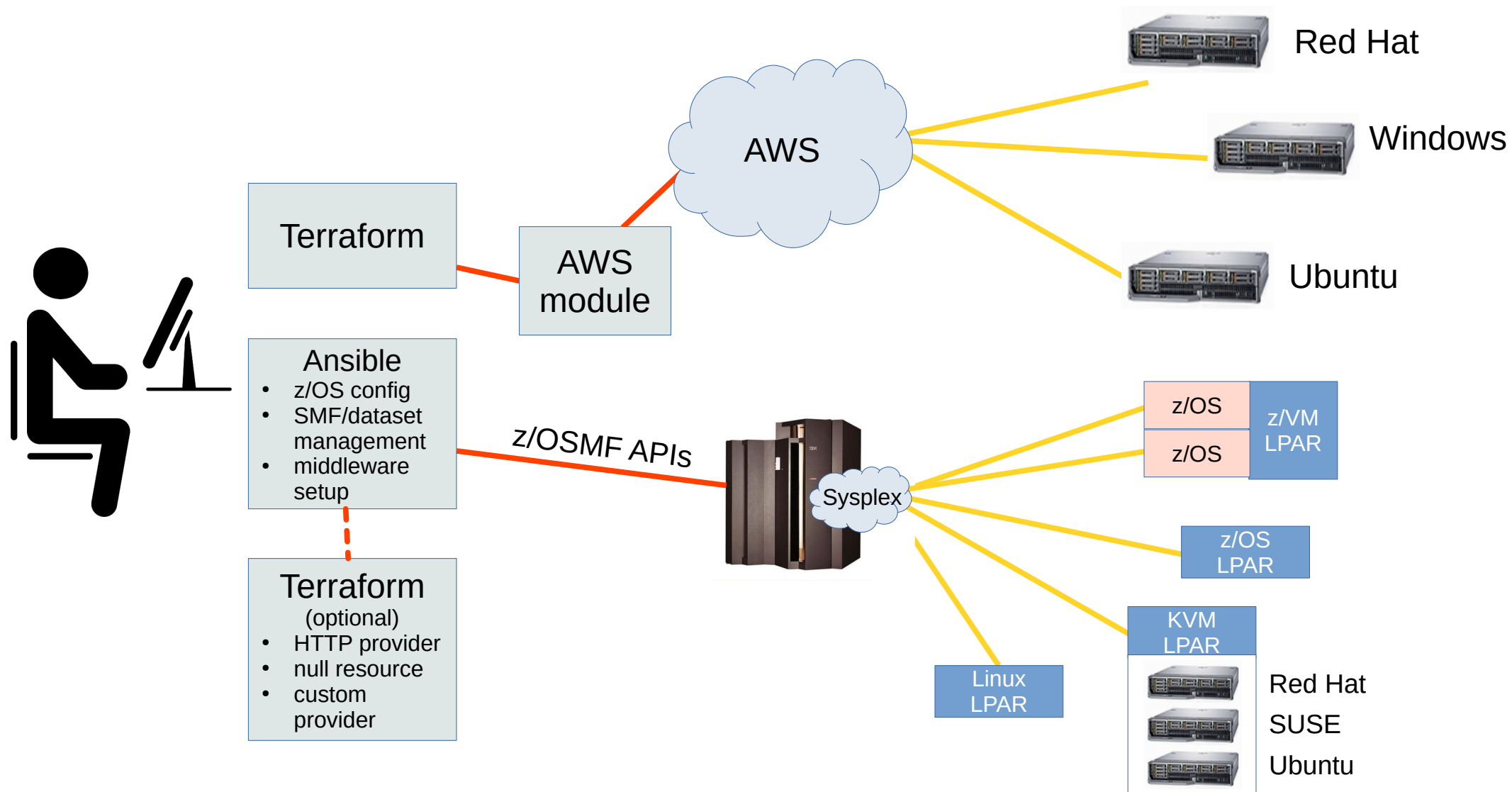
HYBRID APPROACH #1



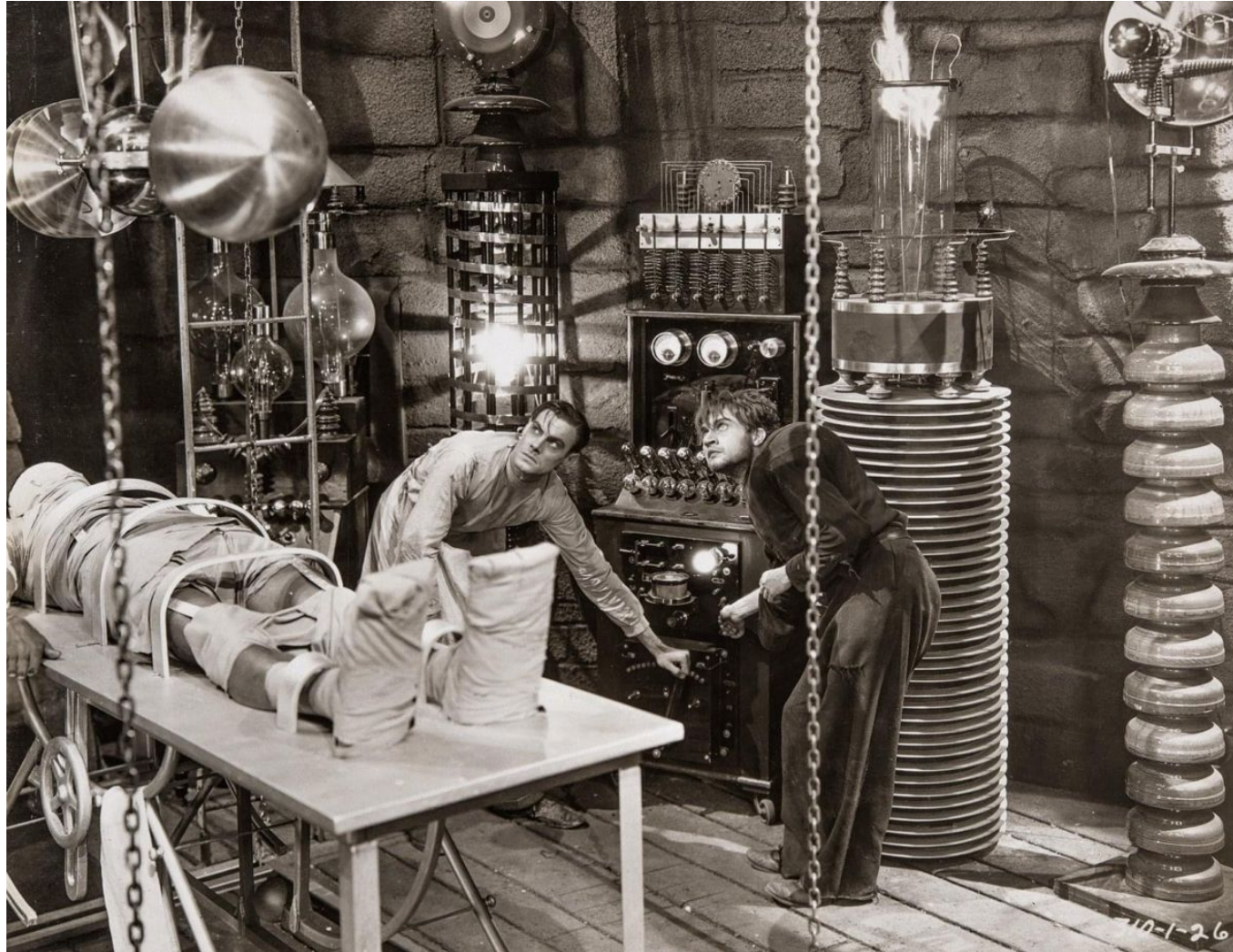
HYBRID APPROACH #2



HYBRID APPROACH #3



LAB Z/OS 44: AUTHENTICATE TO Z/OSMF



WHAT YOU KNOW AND WHAT YOU CAN DO

- You can access z/OSMF with Ansible.

WHAT'S NEXT

- TBD